



SPN Coin

Technical Whitepaper — Version 1.4 · 2026

A Bitcoin-inspired, proof-of-work UTXO blockchain implemented from scratch in Node.js.

Ticker: SPN · Consensus: SHA-256d PoW · Max supply: 21,000,000 · Block time: 10 min

Educational & experimental — no promise of profit

Important notice. This whitepaper is written to be technically accurate and honest. SPN Coin is an educational, experimental blockchain. It is **not** a security or investment product, and this document makes **no promise of profit or returns** and is not an offer to sell or solicit any asset. It describes how the software works, not a financial opportunity. Evaluate SPN Coin as a software-engineering project.

Abstract

SPN Coin is a complete, working implementation of a Bitcoin-style cryptocurrency written entirely in Node.js. It demonstrates — honestly and without shortcuts — how the core mechanisms of a proof-of-work blockchain actually work: the UTXO ledger, real 256-bit proof-of-work, cumulative-work fork choice, peer-to-peer gossip networking, a Stratum mining pool, and a token layer for issuing assets on top of the base chain. This paper explains the design and the reasoning behind each component and is deliberately transparent about the project's limitations.

1. Introduction

1.1 Motivation

Most people who use cryptocurrencies never see how they work internally. Many "educational" blockchain projects fake the hard parts — using a difficulty rule that just counts leading zeros, skipping real fork choice, or omitting a genuine P2P layer. SPN Coin was built to avoid those shortcuts and implement each mechanism the way a production chain would, so the code itself is a faithful teaching reference.

1.2 What SPN Coin is — and is not

It is: a faithful, from-scratch implementation of Bitcoin-style consensus and networking, with a modern web interface, a token layer, and production-grade software-engineering practices (tests, CI/CD, security hardening). **It is not (yet):** a large, live distributed network with real economic value.

1.3 Purpose & vision

SPN Coin is not intended to be "just another coin." Its guiding purpose — inherited from the project's origin — is to bring fast, low-cost, blockchain-based payments to the **transportation industry**: maritime, rail, and road freight, and the businesses around them (shipping lines, railways, trucking and logistics firms, and port and terminal operators). Traditional cross-border settlement in this sector still relies on slow, expensive legacy rails; the project's long-term vision is to offer a transparent, programmable settlement layer as an alternative.

This vision is a **direction of development, not a current deployment**. The project began as Sepanta Token, a transport-focused token, and SPN Coin continues that mission on an independent, purpose-built proof-of-work chain. Real adoption in the transportation industry — integrations, pilots, and partners — is a goal the project is working toward, and this document does not claim it is already in production. Honest progress toward that goal, rather than promises, is how the project intends to earn trust.

2. System Architecture

2.1 Consensus layer

- **Proof-of-Work.** Full 256-bit integer targets (not a "count the zeros" approximation).
- **Cumulative-work fork choice.** The chain with the most total work wins — the correct Nakamoto rule.
- **Median-Time-Past (BIP-113).** Block timestamps validated against the median of recent blocks.
- **Difficulty retargeting.** Difficulty adjusts to keep block production near the target interval.

2.2 Ledger layer

- **UTXO model.** Balances are unspent transaction outputs, exactly like Bitcoin.
- **secp256k1 signatures.** Real ECDSA cryptography for transaction signing and verification.
- **Base58Check addresses.** Addresses use the SPN1... (mainnet) / SPNt... (testnet) prefix, derived from HASH160 of the public key.

2.3 Network layer

- **P2P gossip.** Blocks and transactions propagate between nodes; nodes sync the chain from peers.
- **Peer management.** Address manager, reputation scoring, and DNS-seed-based discovery, following Bitcoin Core patterns.

2.4 Application layer

- **Stratum mining pool** (Vardiff, PPLNS) over WebSocket.
- **Token layer** for issuing assets on the base chain, comparable to ERC-20 / TRC-20.
- **REST + WebSocket APIs**, a block explorer, wallet, and dashboards.

3. Monetary Policy (Simulated)

These parameters mirror Bitcoin's design for educational fidelity. They are a **simulation** of monetary policy, not an economic claim.

Parameter	Value
Ticker	SPN
Smallest unit	1 satoshi = 10^{-8} SPN
Initial block reward	50 SPN
Halving interval	210,000 blocks
Target block time	10 minutes
Maximum supply	21,000,000 SPN (asymptotic)
Consensus	SHA-256d Proof-of-Work
Minimum fee	1,000 satoshi (0.00001 SPN)
Fee model	Size-based, ~10 satoshi/byte

The block reward halves every 210,000 blocks, so the total supply approaches but never exceeds 21,000,000 SPN – the same emission curve as Bitcoin.

Fair launch – zero premine. The genesis block creates no spendable coins. There is no founder allocation and no pre-mined supply: every SPN in existence is produced by proof-of-work mining from block 1 onward, under the same rules for everyone. This is enforced in consensus code, not merely promised.

4. Token Layer

- **Shared transport.** Token operations are carried inside base-chain transactions, so tokens inherit the security of the base chain rather than running a separate network.
- **Per-token contract identity.** Each token receives a deterministic, forge-proof identity (contract address, contract ID, fingerprint, checksum) derived from its immutable fields.
- **Rich metadata.** Logo, description, and social links stored on-chain with the token.

- **Operations.** Issue, mint (if mintable), transfer, burn, freeze/unfreeze, and metadata updates — each authorized by the issuer's signature.

5. Security

- **Transport:** TLS, security headers, CSRF protection.
- **Application:** rate limiting, a DDoS guard (per-IP caps, burst auto-ban, global shedding), and an IDS/IPS engine inspecting for SQLi, XSS, path traversal, command injection and scanner tools.
- **Access control:** role-based access control (RBAC) for node administration, scrypt-hashed credentials, AES-256-GCM encryption, TOTP two-factor authentication, and login logout.
- **Client-side keys:** wallet private keys and recovery phrases are generated and signed in the browser and never sent to the server.
- **Assurance:** automated test suite, CI/CD with CodeQL scanning, and a documented dependency-security process.

Note the deliberate two-layer design: the **chain** is permissionless (anyone with a valid key and signature can transact, like Bitcoin), while the **node admin panel** uses RBAC. RBAC governs who can operate a node, not who can use the chain.

6. Project History

The project began under the earlier name **Sepanta Token**, a token issued on an existing smart-contract chain and aimed at the transportation industry. **SPN Coin — the subject of this whitepaper — is a separate, next-generation implementation:** an independent, from-scratch **proof-of-work Layer-1 blockchain** with its own UTXO ledger and monetary policy (fixed maximum supply of 21,000,000, SHA-256d consensus). It carries forward the same mission on purpose-built infrastructure, but it is a distinct technical system from the earlier token — a different chain, supply, and design.

7. Limitations and Honest Assessment

This section is intentionally candid.

- **Network maturity.** A real blockchain derives its security from a large, independent, distributed set of nodes and hash power. Growing that distributed network is ongoing work.
- **Independent audit pending.** While the code applies strong security practices, it has not yet undergone the independent, professional security audit that handling significant real value would require.
- **Trust is built over time.** A durable cryptocurrency requires a distributed network, an active community, liquidity, legal structure, and years of trust-building — none of which code alone can produce.

8. What SPN Coin Is Good For

- **Learning:** a faithful, readable reference for how a Bitcoin-style chain works end to end.
- **Teaching:** a basis for an educational testnet for a small community.
- **Demonstration:** a portfolio-grade example of full-stack and systems engineering — consensus, cryptography, networking, security, and UI.

9. Roadmap (Non-Financial)

The roadmap is expressed in terms of **software and community**, not price or markets:

- **Decentralization.** Bring up independent nodes and a stable public testnet with independent operators.
- **Independent security audit** and a public bug-bounty program.
- **Ecosystem.** Browser-extension and mobile wallets, an SDK, and complete developer documentation.
- **Community.** Grow a group of contributors and learners around the codebase.

10. Conclusion

SPN Coin demonstrates that the core machinery of a Bitcoin-style blockchain — proof-of-work, UTXO accounting, cumulative-work consensus, P2P networking, and a token layer — can be implemented faithfully and completely in Node.js, with professional software-engineering practices around it. It is offered as an educational and portfolio project, described honestly.

This document is for educational and informational purposes only. It is not financial, investment, or legal advice, and it is not an offer to sell or solicit any asset. Any exchange-listing certificate reproduced here is presented as provided by the project and should be verified independently. © 2026 SPN Coin Project.